

Sammy Abdella

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New York City, NY

Computer Scientist

I am a results-driven software engineer with a passion for backend systems, data optimization, and scalable solutions. My background spans financial institutions, web development, and academic research, with a consistent record of improving system efficiency, ensuring compliance, and enabling better decision-making through data.

Areas of Expertise

Software Development • API Optimization • Microservices Architecture • Data Analysis • PostgreSQL Optimization • OAuth2/JWT • Docker & Kubernetes • Scientific Computing • Website Development • Research • Multilingual Communication (English, French, Arabic)

Education

Bachelor of Science in Computer Science | Columbia University – New York City, NY

May 2027

Professional Work History

Web Developer Intern | Columbia College Information Technology

May 2025 – Present

- Migrated 43 Columbia departmental websites from Drupal 7 to Drupal 10, writing migration scripts, refactoring custom modules, handling regression testing, and improving average page load times by ~35% and eliminating 100% of deprecated modules in production
- Integrated a Django, PostgreSQL, and RESTful API backend with a new Columbia admissions website, scaling reliably to 50,000+ applicants per cycle, cutting form submission errors by ~40%, and reducing manual data correction work by ~50%

Research | Columbia University Artificial Intelligence Laboratory

January 2025 – September 2025

- Led the development for gAyl, an AI tool for LGBTQ+ suicide prevention, designed six model-based safety benchmarks to generate treatment plan recommendations and achieved <15% false-negative rate on unsafe-response detection across 1,000+ high-risk prompts
- Trained LLM-based classifiers in PyTorch and Hugging Face Transformers to automate safety evaluation, reducing manual benchmarking time by ~95%
- Publication: <https://doi.org/10.7916/z6vc-d867>

Remote Software Developer Intern | Zitouna Bank

January 2024 – May 2024

- Engineered RESTful APIs using Python/Django to cut response times by 30% and boost transaction throughput by 25%
- Partnered with database teams to tune PostgreSQL performance for real-time, high-volume financial data, reduced p95 query latency by ~35%
- Ensured backend infrastructure aligned with security standards and financial regulations, resulting in a 100% compliance rate during audits

Remote Freelance Software Developer | Fiverr

November 2021 – September 2025

- Assisted 50+ clients in customizing web applications, developing mobile apps, integrating APIs, optimizing website performance, and implementing cybersecurity measures, \$10,000+ total earnings

Technical Skills

Python • C++ • Java • JavaScript • SQL • HTML/CSS • Django • RESTful APIs • OAuth2 • JWT • Docker • Kubernetes • PostgreSQL • API Integration • Web Optimization • Kubernetes • Large Language Models (LLMs) • Jupyter Notebook • Matplotlib • Scientific Computing • PostgreSQL Performance Tuning • SQL Query Optimization • Database Indexing • API Performance Optimization • Microservices Architecture • CI/CD Pipelines • Unit Testing • Regression Testing • Content Management Systems (CMS) • Drupal 7–10 Migrations • Website Performance Optimization • Security & Compliance (Backend Systems)